

现有论文方法

不需要生成数据+训练模型的方法

ARXIV_Towards Training-free Open-world Segmentation via Image Prompt Foundation Models

#DINO #SD #不同提示词的去噪图像差值

输入提示图像和待分割图像，分别提取DINOv2和SD特征融合，对提示图像特征做无监督显著性目标检测提取前景特征提示，然后和输入图像的特征匹配，得到前景背景候选点，送给SAM继续分割。

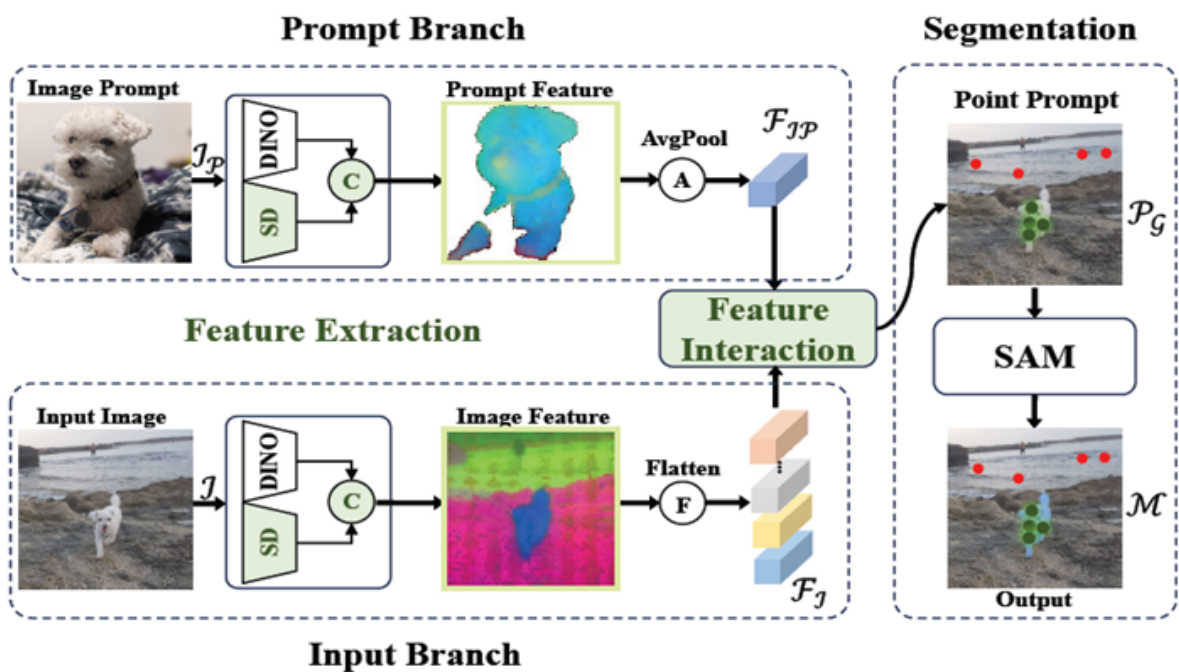


Fig. 3 The framework of our proposed IPSeg framework. Importantly, all parameters in the network remain frozen, eliminating the need for additional training. The green point in $\mathcal{P}_G$ represents the positive point prompts sent to SAM, while the red point represents the negative point prompts sent to SAM.

ARXIV_Diffusion Model is Secretly a Training-free Open Vocabulary Semantic Segmenter

#attention得到粗略图 #CRF优化 #k

不需要训练模型的DiffSegmenter，输入图像和类别集合或者一句话给SD，通过去噪UNet网络的交叉注意力机制得到粗略图，通过自注意力机制优化，设计文本提示和类别过滤机制，输入图像给BLIP生成CAPTION，通过CLIP初步判断类别关键词。对弱监督语义分割任务，还要用CRF优

化**得到伪真值，然后训练语义分割模型。

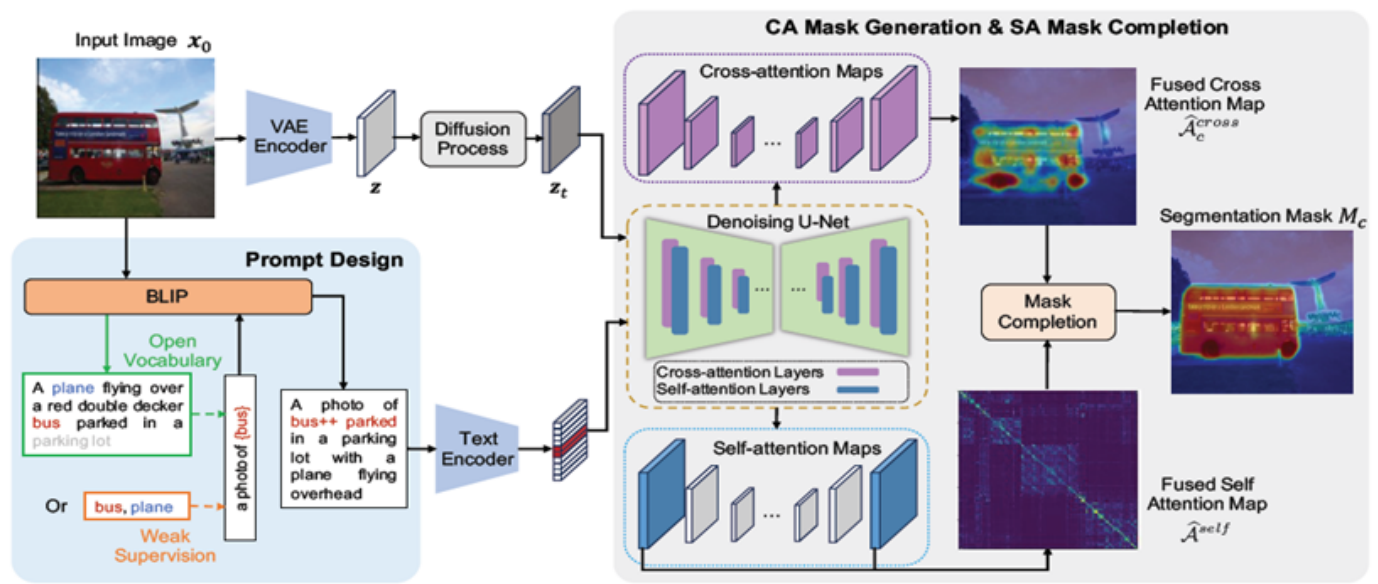


Figure 2: **Overview of the proposed DiffSegmter.** An input image and enhanced candidate class tokens by the BLIP-based prompt design module are fed into an off-the-shelf pre-trained conditional latent diffusion model. The fused cross-attention maps produced by the denoising U-Net are treated as the initial segmentation score maps, which is further refined and completed by the fused self-attention maps of the U-Net. Note that the parameters of all the involved models are frozen without any tuning.

Factorized Diffusion Architectures for Unsupervised Image Generation and Segmentation

#MaskGenerator

#掩码分支去噪

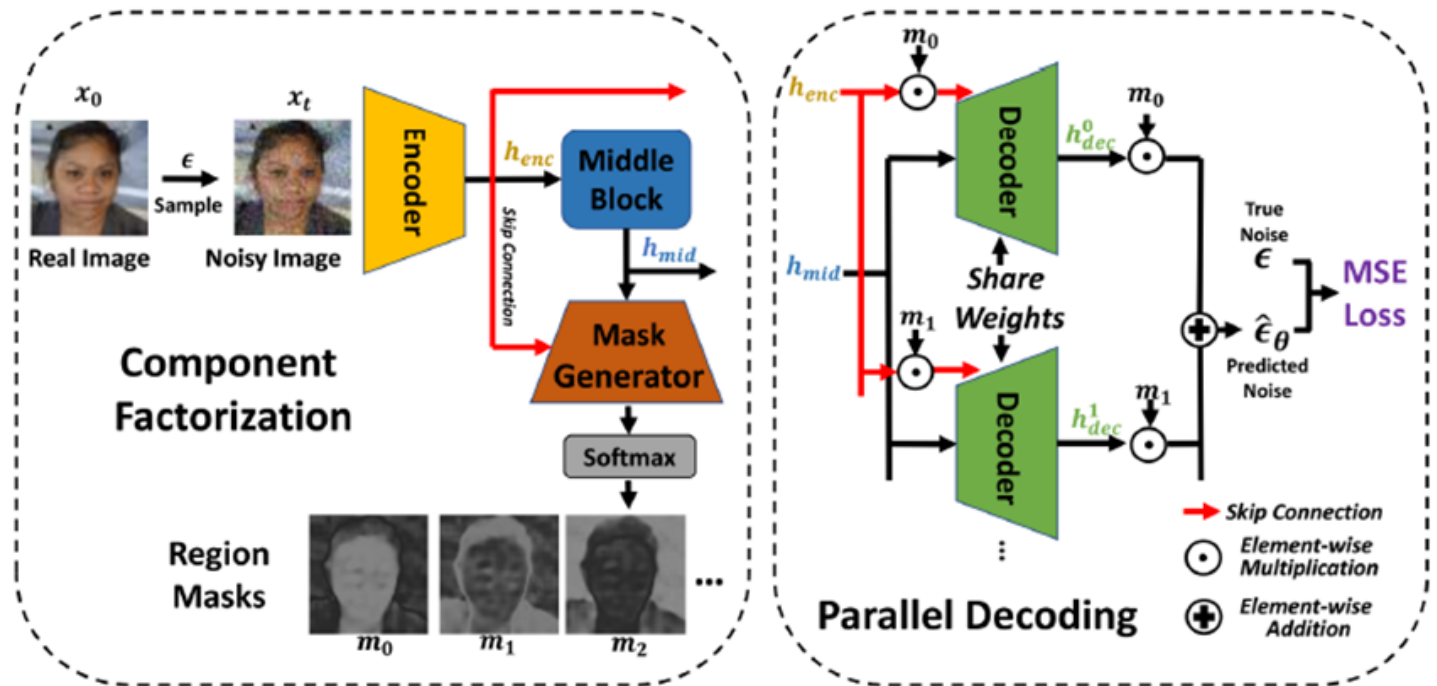


Figure 2. **Factorized diffusion architecture.** Our framework restructures the architecture of the neural network within a DDPM [25] so as to decompose the image denoising task into parallel subtasks. All modules are end-to-end trainable and optimized according to the same denoising objective as DDPM. **Left: Component factorization.** An **Encoder**, equivalent to the first half of a standard DDPM U-Net architecture, extracts features h_{enc} . A common **Middle Block** processes **Encoder** output into shared latent features h_{mid} . Note that **Middle Block** and h_{mid} exist in the standard denoising DDPM U-Net by default. We draw it as a standalone module for a better illustration of the detailed architectural design. A **Mask Generator**, structured as the second half of a standard U-Net receives h_{mid} as input, alongside all encoder features h_{enc} injected via skip connections to layers of corresponding resolution. This later network produces a soft classification of every pixel into one of K region masks, $m_0, m_1, \dots, m_K$. **Right: Parallel decoding.** A **Decoder**, also structured as the second half of a standard U-Net, runs separately for each region. Each instance of the **Decoder** receives shared features h_{mid} and a masked view of encoder features $h_{enc} \odot m_i$ injected via skip connections to corresponding layers. Decoder outputs are masked prior to combination. Though not pictured, we inject timestep embedding t into the **Encoder**, **Mask Generator**, and **Decoder**.

ARXIVDiffuse, Attend, and Segment Unsupervised Zero-Shot Segmentation using Stable Diffusion

#attention得到粗略图

输入一张图片实现zero shot分割，主要依赖于图片输入给SD的自注意力，首先对**自注意力图聚集**，然后基于**相似性和KL散度**进行合并，非极大抑制

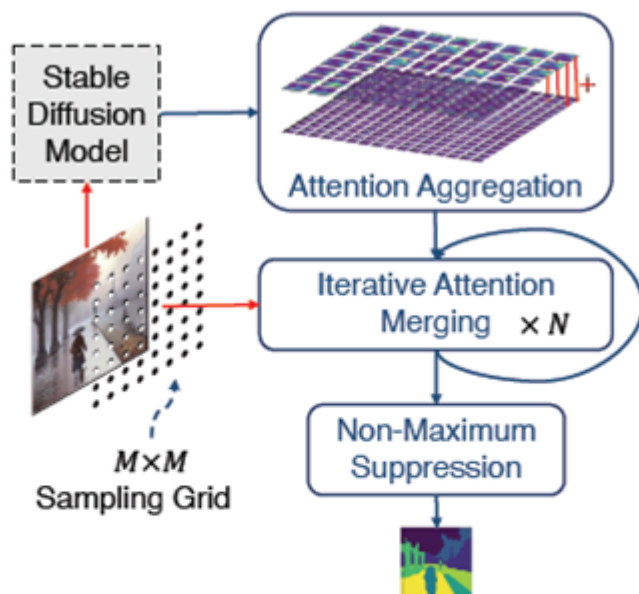


Figure 1. Overview of DiffSeg. DiffSeg is an unsupervised and zero-shot segmentation algorithm using a pre-trained stable diffusion model. Starting from $M \times M$ anchor points, DiffSeg iteratively merges self-attention maps from the diffusion model for N iterations to segment any image without any prior knowledge and external information.

需要生成数据+训练模型的方法

ARXIV_Diffusion Models for Zero-Shot Open-Vocabulary Segmentation

#DINO

#SD

#原型

#k

分割一个图像，对每个类别**生成一批图像**，每个图像**提取DINO、SD等特征**，通过**无监督方法CutLER**提取**前景背景特征原型**。同样对输入图像提取特征，与前景背景特征**原型计算相似度**实现

分割。先用CLIP初步挑选类别，用ChatGPT排除背景类别干扰？

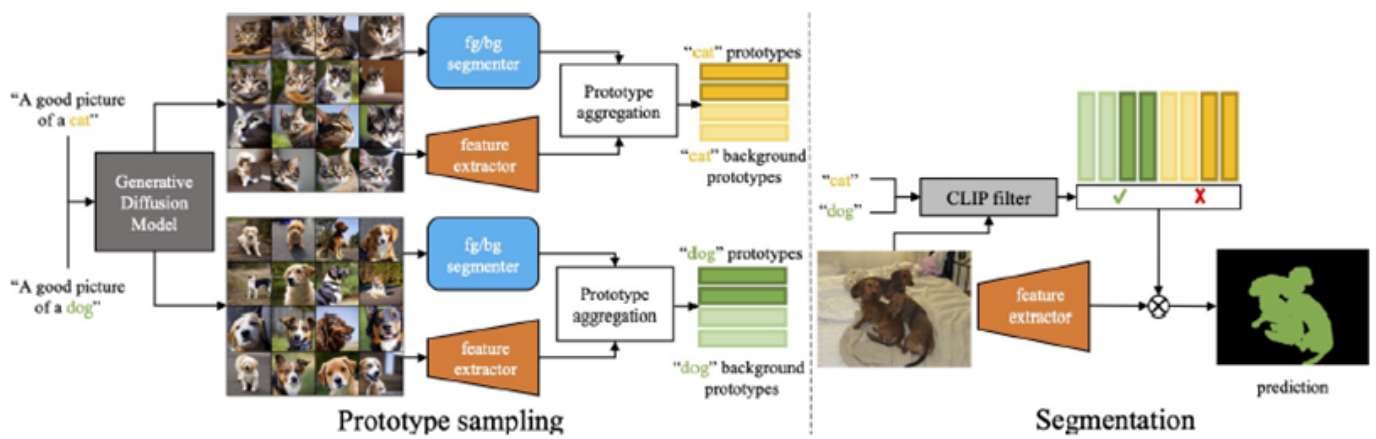


Figure 1: **OVDiff overview**. There are two stages: during prototype sampling, text queries are used to sample a set of support images which are further processed by a feature extractor and a segmenter forming positive and negative or background prototypes. During the segmentation stage, image features are extracted and compared against prototypes, where all background prototypes map to the background class. The CLIP filter removes irrelevant prototypes based on global image contents.

ICCV2023_Open-vocabulary Object Segmentation with Diffusion Models

#SD生成

这里的所谓OPEN，指能对SD能生成的任意类别目标，同时能生成MASK，不限制语义类别。先通过SD生成部分类别图片，然后用MASK RCNN推理得到MASK？通过一个Transformer decoder（不是SD的交叉注意力特征）建立文本特征和SD特征之间的关系+MLP训练网络，然后能在其他Unseen类别上生成MASK？

通过文本提示只生成1-2个目标。首先实验训练grounding模块用于生成SD生成图的MASK，然后实验训练语义分割网络，用MaskFormer做backbone，评测open segmentation性能。

代码 https://lipurple.github.io/Grounded_Diffusion/

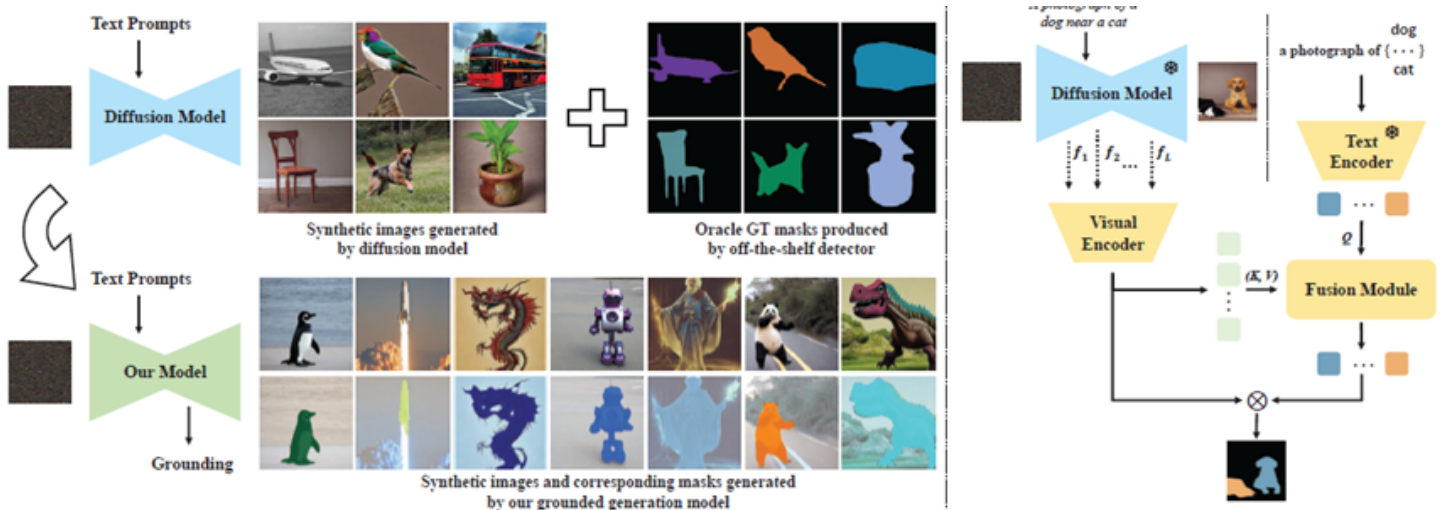


Figure 2: **Overview**. The left figure shows the knowledge induction procedure, where we first construct a dataset with synthetic images from diffusion model and generate corresponding oracle groundtruth masks by an off-the-shelf object detector, which is used to train the open-vocabulary grounding module. The right figure shows the architectural detail of our grounding module, that takes the text embeddings of corresponding entities and the visual features extracted from diffusion model as input, and outputs the corresponding segmentation masks. During training, both the diffusion model and text encoder are kept frozen.

CVPR2023_Open-Vocabulary Panoptic Segmentation with Text-to-Image Diffusion Models

#Caption生成 #Unet语义 #CLIP相似度 #k

输入图片，**生成隐含的Caption**，然后作为条件输入给Stable Diffusion，进而**提取UNet语义特征**（需要在有标注的基类数据集上训练。通过计算每个Region图像特征和CLIP文本**特征相似性**，可以泛化到Novel类）。

代码 <https://github.com/NVlabs/ODISE>

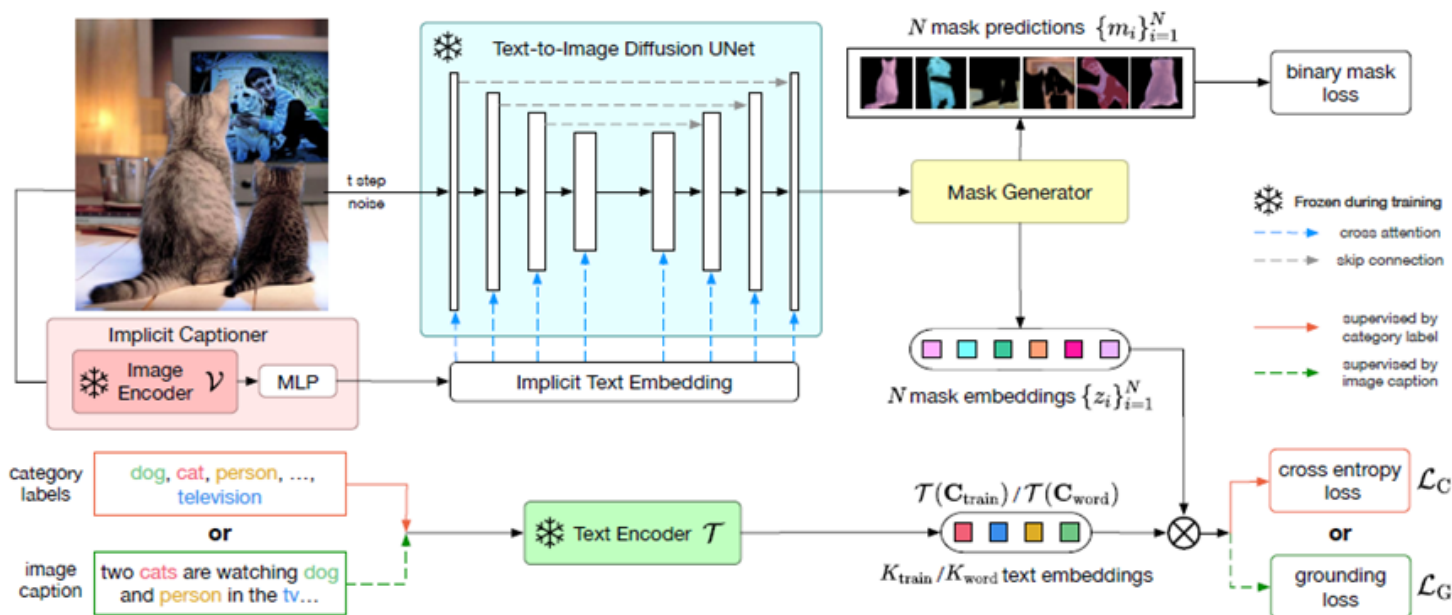


Figure 2. ODISE Overview and Training Pipeline. We first **encode the input image into an implicit text embedding** with an implicit captioner (image encoder $\mathcal{V}$ and MLP). With the image and its implicit text embedding as input, we **extract their diffusion features** from a frozen text-to-image diffusion UNet (Sec 3.3). With the UNet’s features, a **mask generator predicts class-agnostic binary masks** and their associated mask embedding features (Sec 3.4). We **perform a dot product** between the mask embedding features and the text embeddings of training category names (red box) or the nouns of the image’s caption (green box) to categorize them. The similarity matrix for mask classification is supervised by either a cross entropy loss with ground truth category labels (red solid path), or via a grounding loss with the paired image captions (green dash path) (Sec 3.5).

ICCV2023DiffuMask Synthesizing Images with Pixel-level Annotations for Semantic Segmentation Using Diffusion Models

#CRF优化 #SD #attention得到粗略图

- 通过**SD的交叉注意力**得到注意力图，然后**通过AffinityNet网络学习自适应的二值化**，交叉验证去掉噪声，**CRF**，用VOC上训练过的**MaskFormer**推理作为伪真值评测性能。Open vocabulary zero shot达到**SOTA**
- 提示语言用WIKI上多个子类，从真实语言中检索最相关的。采用拼图、模糊、遮挡、仿射变换等数据增强方法。只生成一个目标？

代码 <https://weijawu.github.io/DiffusionMask/>

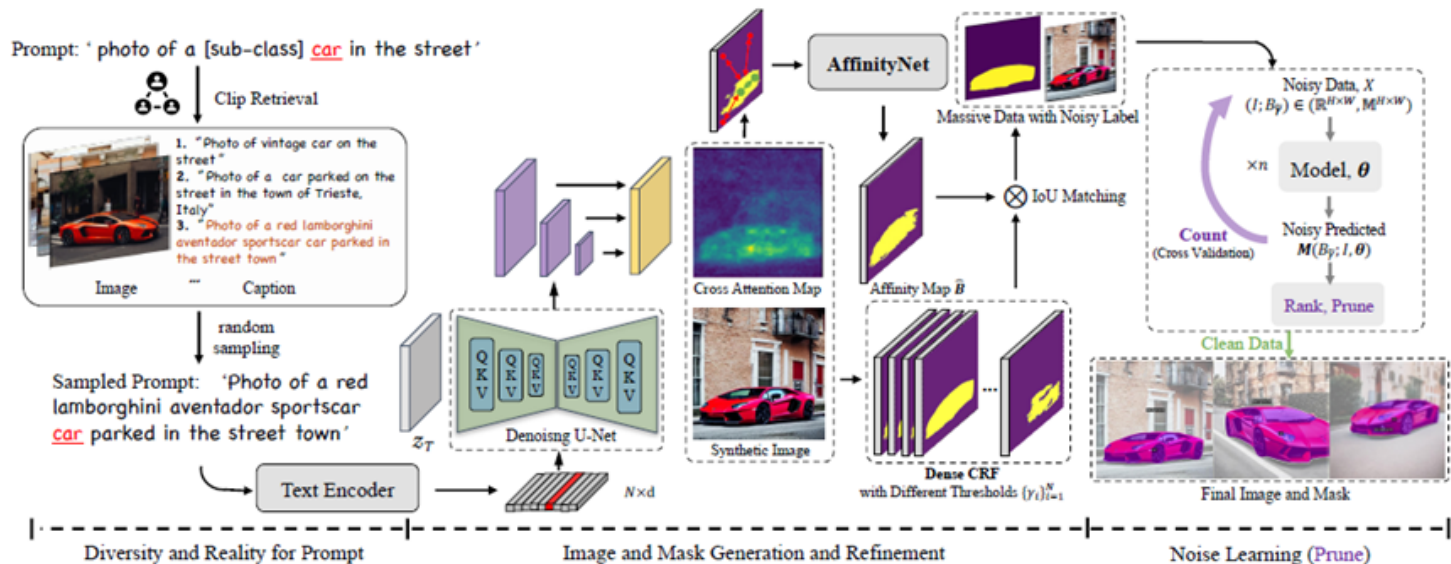


Figure 4 – Pipeline for DiffuMask with a prompt: ‘Photo of a [sub-class] car in the street’. DiffuMask mainly includes three steps: 1) Prompt engineering is used to enhance the diversity and reality of prompt language (Sec. 3.4). 2) Image and mask generation and refinement with adaptive threshold from AffinityNet (Sec. 3.2). 3) Noise learning is designed to further improve the quality of data via filtering the noisy label (Sec. 3.3).

NIPS2023Dataset diffusion Diffusion-based synthetic dataset generation for pixel-level semantic segmentation

#Caption生成 #attention得到粗略图 #k

通过ChatGPT、COCO数据集的Caption，VOC数据集生成Caption，得到提示语言，并加上类别单词确保每个类别都生成了（看来提示语言非常重要？分析在Table2）。通过SD的自注意力和交叉注意力得到初步分割，标记为前景、背景、未知，训练语义分割网络。

生成的syncVOC上性能超过DiffuMask（只能生成单个目标？），生成的syncCOCO上性能达到真实图像全监督方法

代码 <https://github.com/VinAIResearch/Dataset-Diffusion>

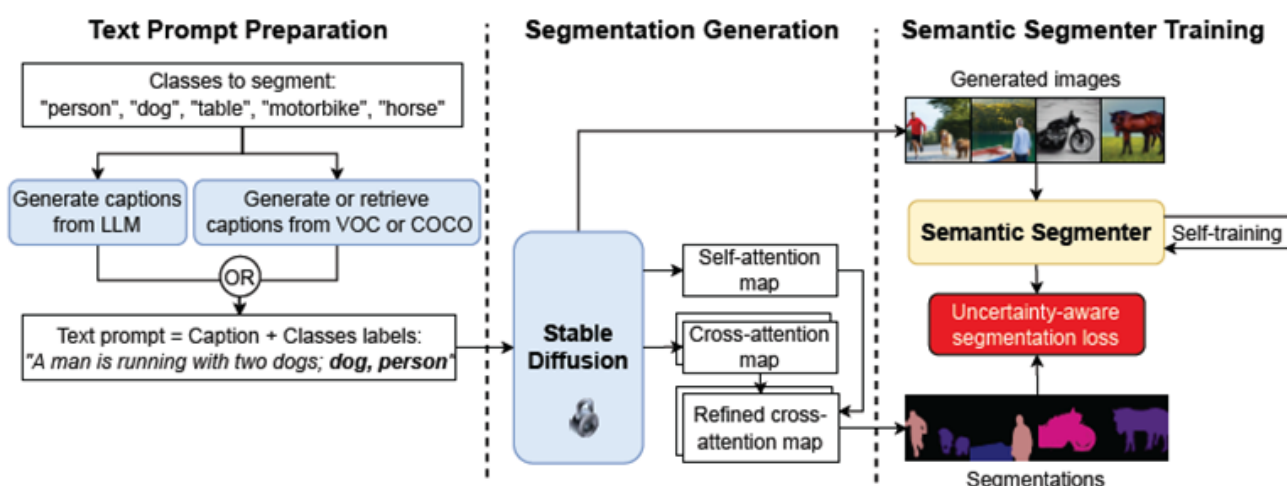
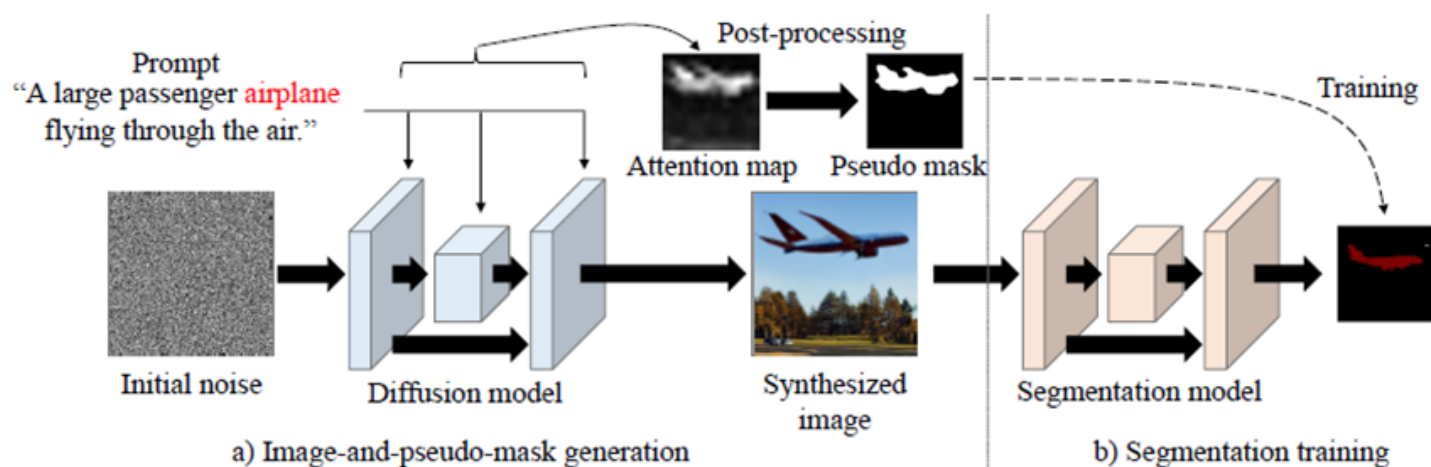


Figure 2: Three stages of Dataset Diffusion. In the first stage, the target classes are provided, and text prompts are generated using language models such as ChatGPT [48]. Real captions (for COCO) or image-based captions (for VOC) can also be used for prompt generation to ensure standard evaluation. The **text prompts** are then augmented with the **target class labels** to avoid missing objects. In the second stage, given the augmented text prompt, a frozen Stable Diffusion [5] is employed to **generate an image and its self- and cross-attention maps**. The cross-attention map for each target class is refined using the self-attention map to match the object’s shape. Finally, the generated images and corresponding semantic segmentations are used to train a semantic segmenter with uncertainty-aware loss and the self-training technique.

ARXIVAttention as Annotation Generating Images and Pseudo-masks for Weakly Supervised Semantic Segmentation with Diffusion

#CRF优化 #attention得到粗略图 #SD生成 #I

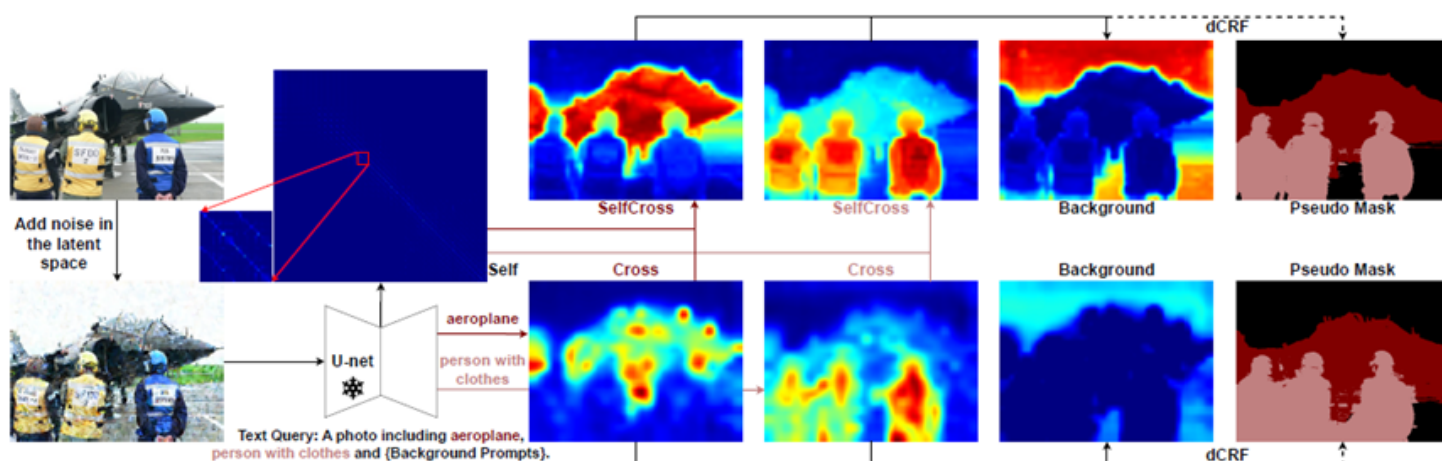
通过SD的交叉注意力和自注意力分割，[CRF](#)优化，生成大量图片和伪真值训练语义分割网络



ARXIVFrom Text to Mask Localizing Entities Using the Attention of Text-to-Image Diffusion Models

#attention得到粗略图 #CRF优化 #k

通过SD的交叉注意力和自注意力分割，[CRF](#)优化



ARXIV_Adapting diffusion towards unsupervised object discovery

#attention得到粗略图 #CLIP相似度

只分割前景背景，和显著性检测差不多。首先通过提示语句生成大量图像，基于交叉注意力和自注意力做类似[GraphCut](#)全局优化得到MASK，训练SD分割网络。测试时输入图像通过CLIP得到类

别，送给SD作为条件，通过SD分割网络输出结果。

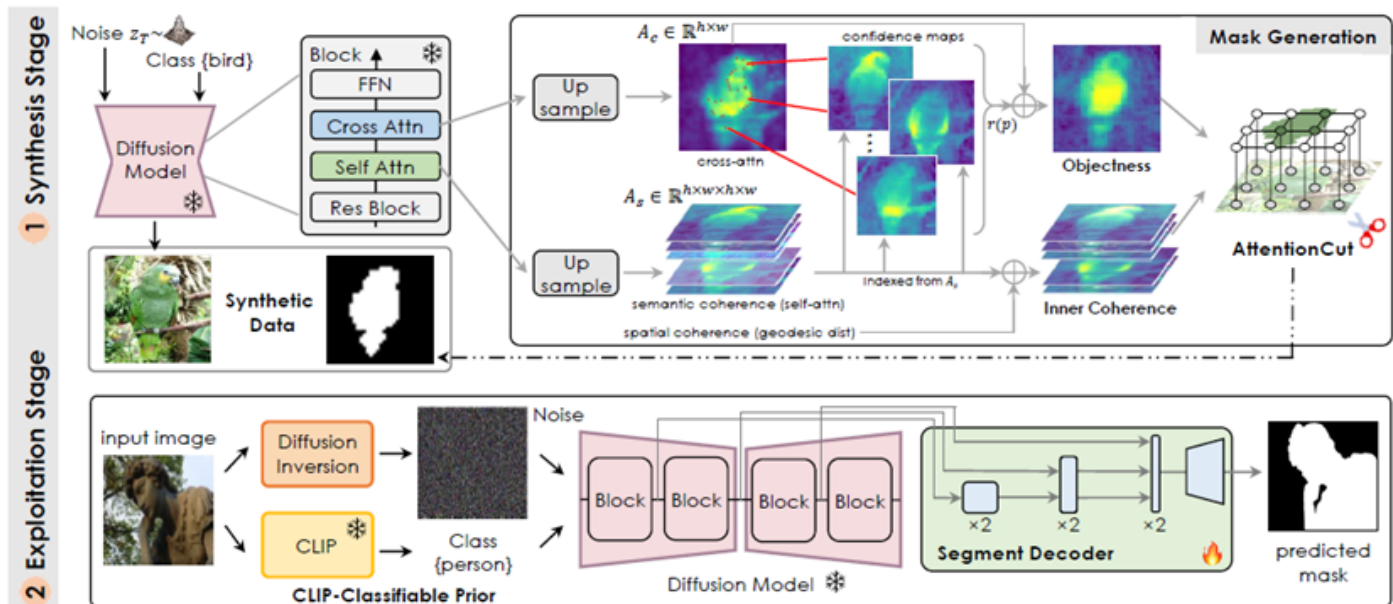


Figure 2: We use pre-trained diffusion models to synthesize extensive data, helping the model training, then evaluated on real images. (1) **The Synthesis Stage.** We synthesize free image-mask pairs, where mask generation is solved leveraging cross- and self- attention by AttentionCut. (2) **The Exploitation Stage.** We extract knowledge using diffusion inversion, then only one lightweight decoder is trained for object discovery on synthetic data.

WACV2024FOSSIL Free Open-Vocabulary Semantic Segmentation Through Synthetic References Retrieval

#k

训练时，通过交叉注意力DAAM生成每个单词激活区域，与DINOv2特征相乘提取特征。推理时，每个类别文本检索类似的N个文本特征对应视觉特征，聚类得到特征原型，输入图像通过DINOv2提取特征，无监督方法OpenCut分割，得到区域特征和特征原型匹配。

可以借鉴应用到分割任务的方法

ICLR2023_DiffEdit Diffusion-based semantic image editing with mask guidance

#Caption差值 #去噪结果差值

【不同文本描述去噪结果的差值】

将原图片加噪，接着**分别**使用描述原图片的**原文本和目标文本的去噪结果**（是去噪结果，不是基于注意力图）的**差值**来得到一个掩码（思路可以用于分割？）之后再使用DDIM的确定性加噪方

式，将图片加到一定程度后，使用类似前面我们提到过的方式分别处理掩码区域和背景区域。

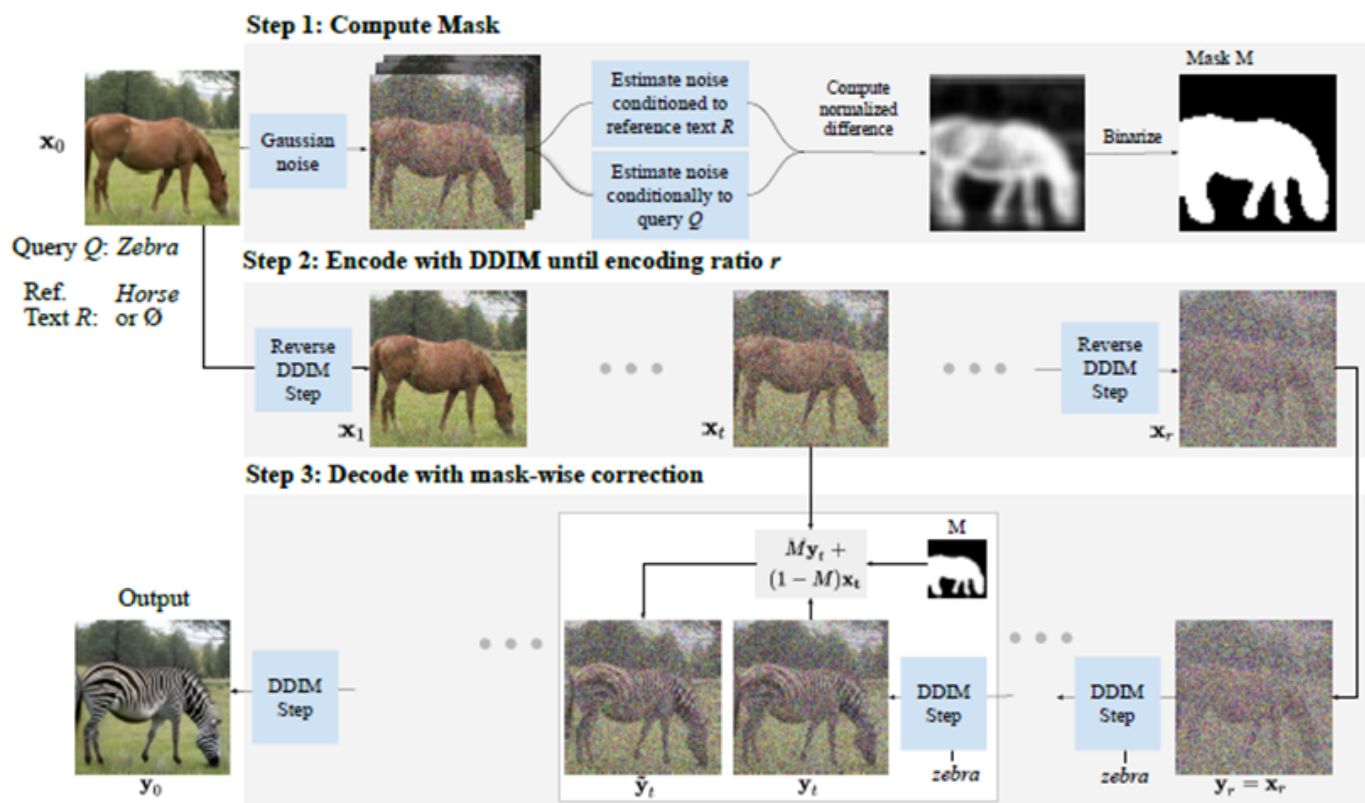


Figure 2: The three steps of DIFFEDIT. **Step 1:** we add noise to the input image, and denoise it: once conditioned on the query text, and once conditioned on a reference text (or unconditionally). We derive a mask based on the difference in the denoising results. **Step 2:** we encode the input image with DDIM, to estimate the latents corresponding to the input image. **Step 3:** we perform DDIM decoding conditioned on the text query, using the inferred mask to replace the background with pixel values coming from the encoding process at the corresponding timestep.

ARXIV_Text-to-Image Diffusion Models are Zero-Shot Classifiers

#去噪结果差值 #与原图差值 #k

将类别作为条件输入给SD和Imagen做去噪，计算去噪重建结果和原图的差异，对不同时间点的差异加权求和，输出差异最小的类别，实现分类。

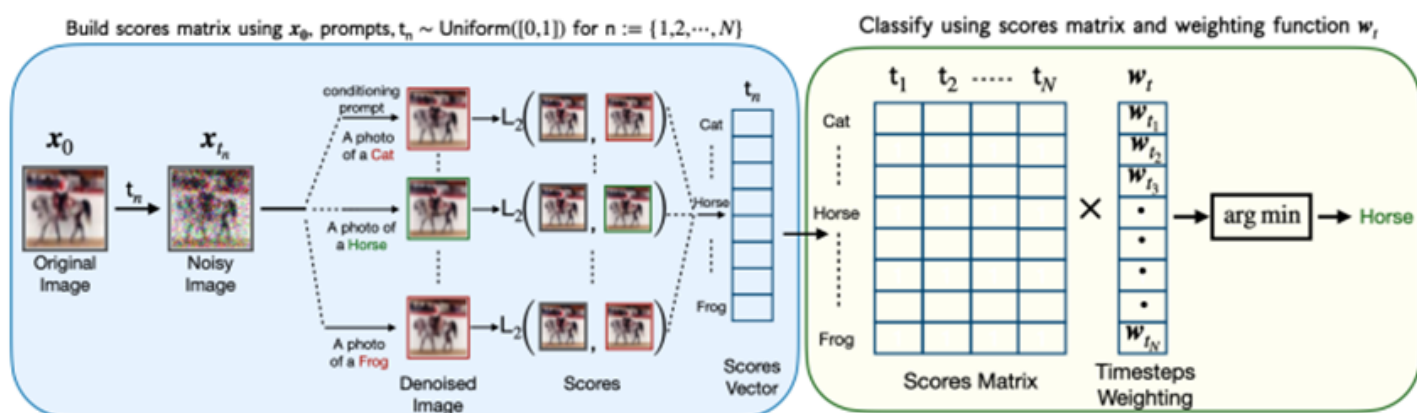


Figure 1: **Zero-Shot Classification using Diffusion Models.** We first compute denoising scores for each label prompt across multiple time-steps to generate a scores matrix. We then classify an image by aggregating the scores for each class using a weighting function over the time-steps. The image is assigned the class with the minimum aggregate score. In Section 3.1, we discuss how efficiency can be improved only computing a subset of the full scores matrix.

ICCV2023_Your Diffusion Model is Secretly a Zero-Shot Classifier

#噪声差值

将类别作为条件输入给SD做去噪，计算**估计的噪声**和**添加的噪声差异**做分类。

代码 diffusion-classifier.github.io/

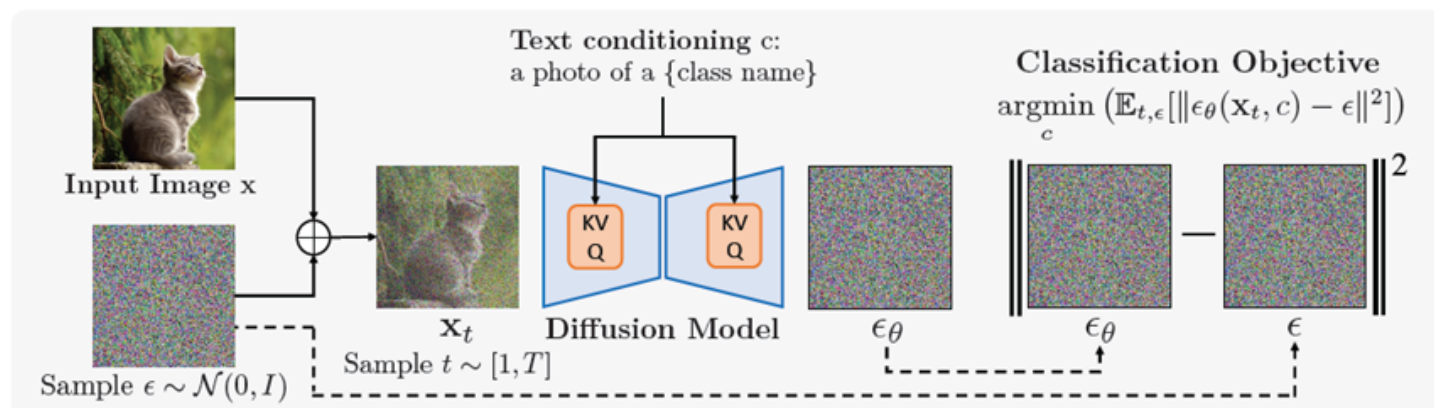


Figure 1. Overview of our Diffusion Classifier approach: Given an input image x and a set of possible conditioning inputs (e.g., text for Stable Diffusion or class index for DiT), we use a diffusion model to choose the one that best fits this image. Diffusion Classifier is theoretically motivated through the variational view of diffusion models and uses the ELBO to approximate $\log p_\theta(x | c)$. Diffusion Classifier chooses the conditioning c that best **predicts the noise added to the input image**. Diffusion Classifier can be used to extract a zero-shot classifier from Stable Diffusion and a standard classifier from DiT without any additional training.

ARXIV_Prompt-to-Prompt Image Editing with Cross-Attention Control

#attention得到粗略图

通过不同类别提示词控制去噪，从UNet的**注意力图**得到出部分分割MASK，然后控制编辑图像

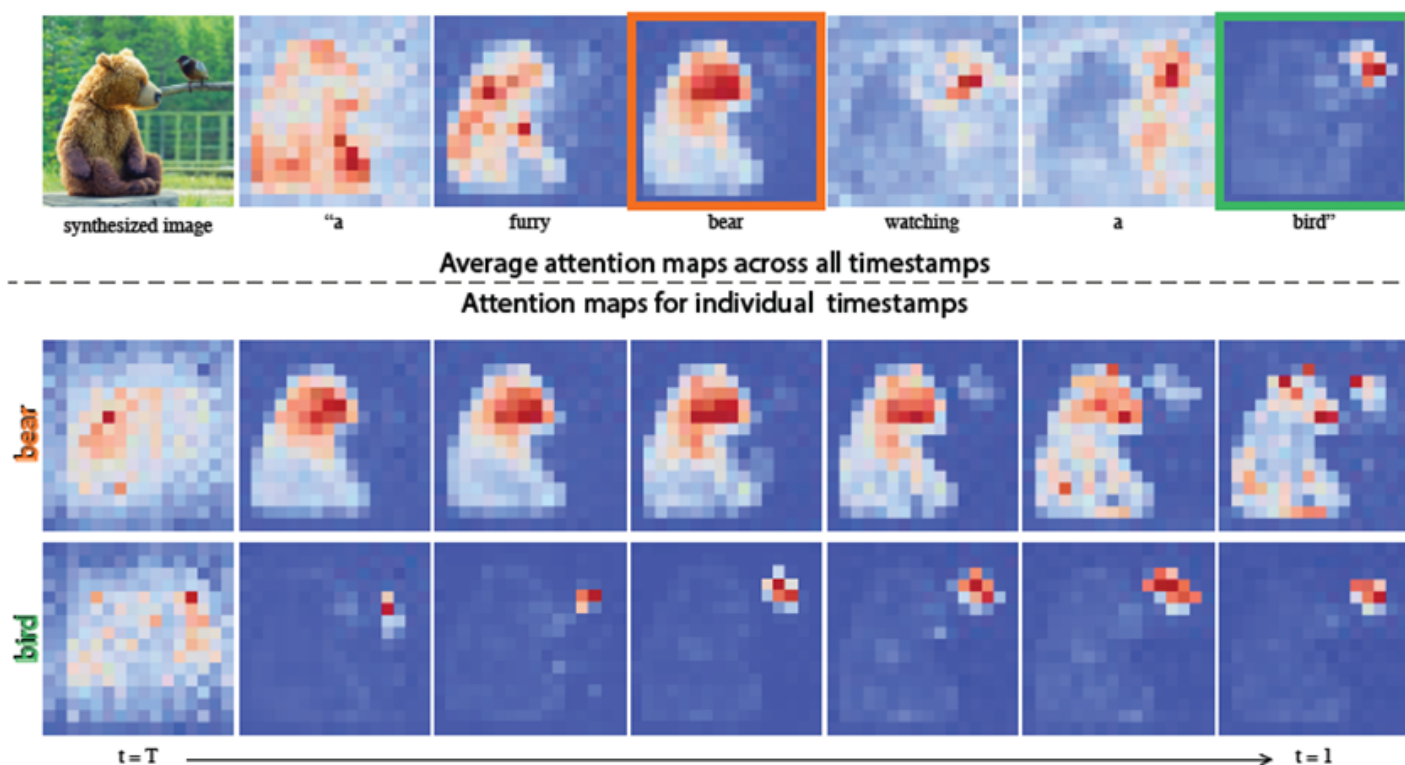


Figure 4: Cross-attention maps of a text-conditioned diffusion image generation. The top row displays the average attention masks for each word in the prompt that synthesized the image on the left. The bottom rows display the attention maps from different diffusion steps with respect to the words "bear" and "bird".

ARXIVBeyond Generation Harnessing Text to Image Models for Object Detection and Segmentation

#Caption生成 #k

用简单提示生成简单图像分割（怎么分割？）出目标，通过真实图像（VOC COCO）生成CAPTION去掉目标得到背景文字（改为CHATGPT更好？）生成背景图像。目标背景通过CLIP排除低质量的，4个目标随机粘贴到背景上。在0,1,10shot（指用多少张真实图片的背景）上评测，性能比真实数据全监督方法还是差很多？。

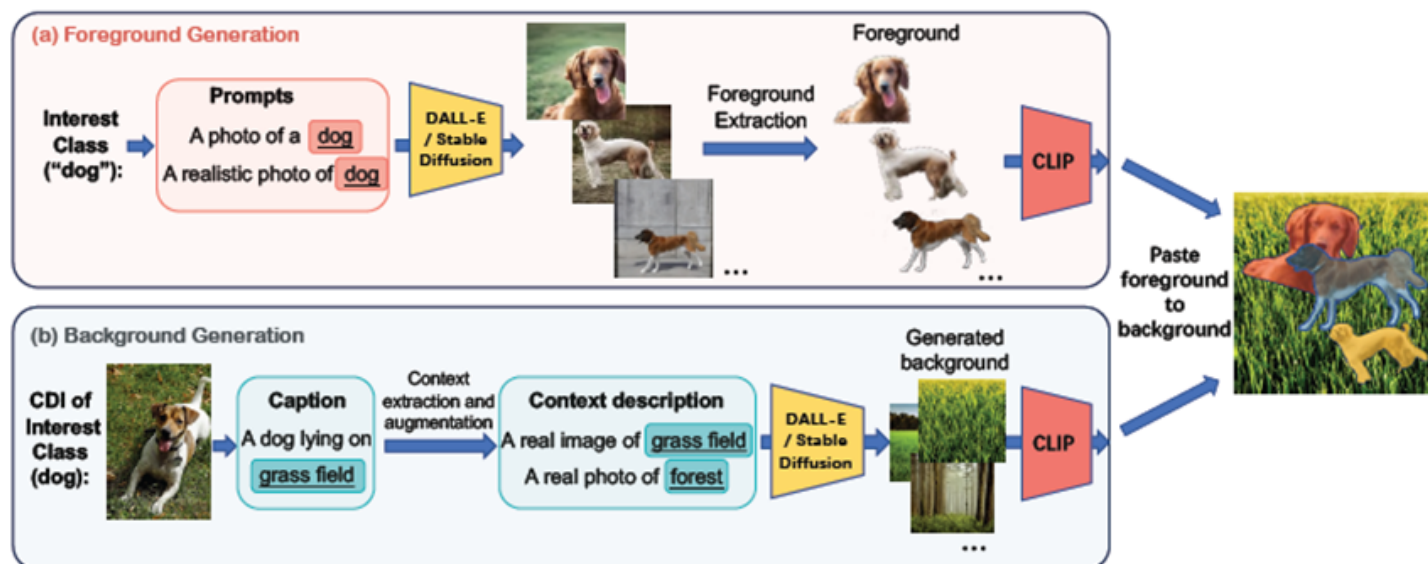


Figure 2: (a) **Foreground generation:** (top row, §3.1) verbalizes class name into templates understandable by T2I models [51, 56], which synthesize desired foreground images with easy-to-separate backgrounds. Off-the-shelf foreground/background segmentation methods are then used to extract foreground segments from foreground images. (b) **Background generation:** (bottom row, §3.2) an image captioning method (e.g. SCST [53]) captions user-provided images (CDIs). Context words (e.g. “grass field”) are extracted and the augmented caption is feed into T2I to generate background images. (c) CLIP [50] is used (§3.3) to maintain the quality of both foregrounds and backgrounds, as well as ensure that the generated images do not have unwanted class. (d) Finally, we composite (§3.4) the foreground segments and background images to obtain synthetic images with accurate labels.

ARXIVDatasetDM Synthesizing Data with Perception Annotations Using Diffusion Models

#attention得到粗略图 #Unet语义 #k

通过SD的交叉注意力和UNET特征训练分割模型，需要少量标注。

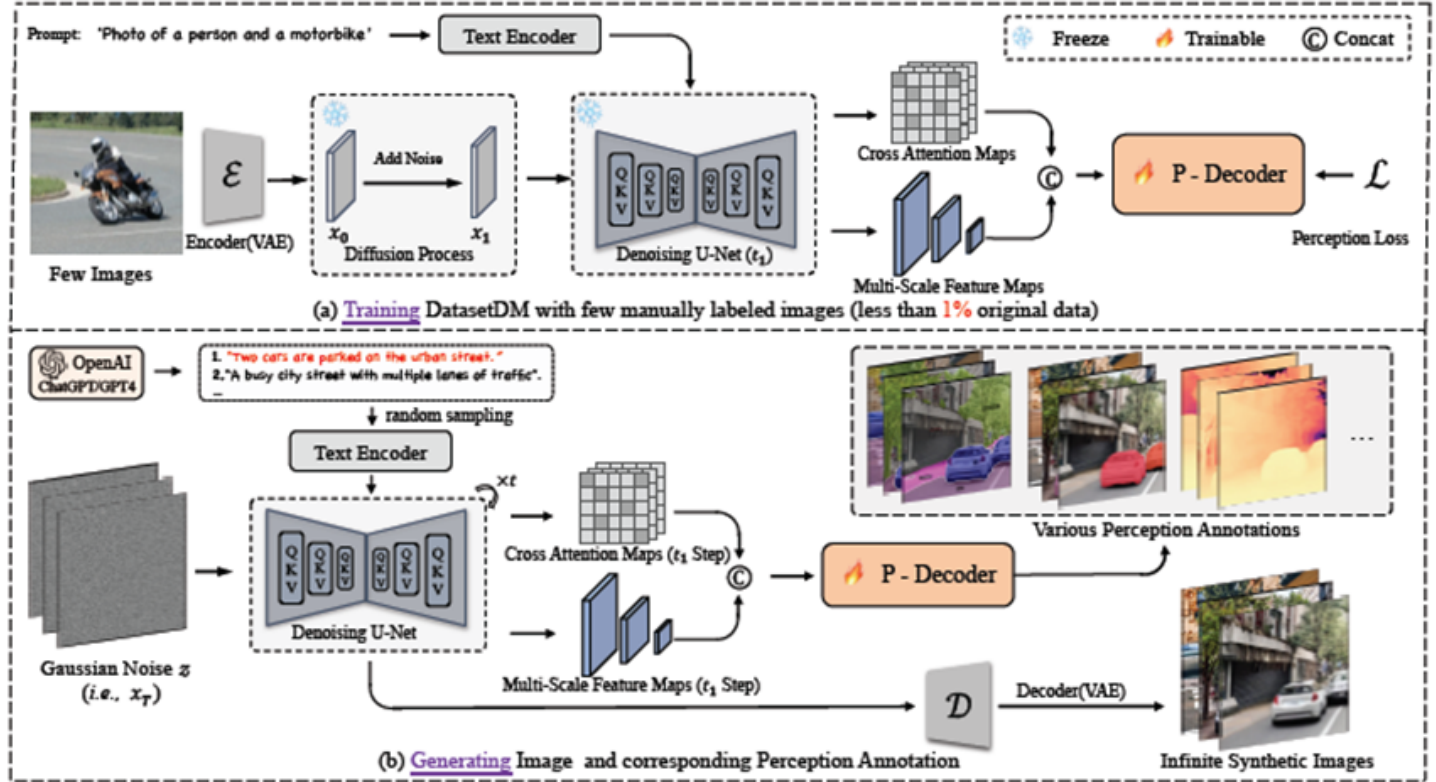
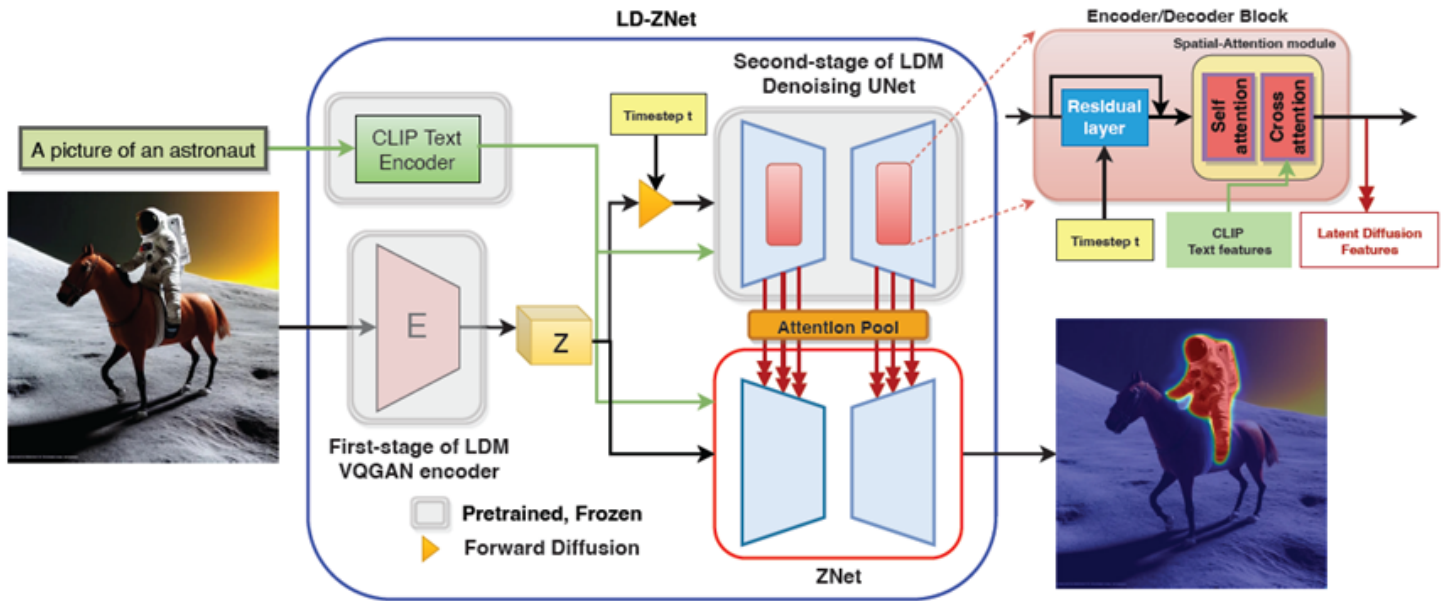


Figure 2: The overall framework of DatasetDM. DatasetDM consists of two main steps: 1) Training. Using diffusion inversion to extract the latent code from a small amount of data and then train the perception decoder. 2) Text-guided data generation. A large language model such as GPT-4 is utilized to prompt infinite and diverse data generation for various downstream tasks.

ICCV2023LD-ZNet A Latent Diffusion Approach for Text-Based Image Segmentation

#Z特征

需要输入文本提示和图片做分割，训练需要MASK。提出的ZNet直接基于Z特征和文本特征做分割，LD-ZNet在此基础上加了SD的特征。在3.2.1节首先提取特征有监督训练分析哪些block哪些时间特征有效。



ARXIVDFormer Diffusion-guided Transformer for Universal Image Segmentation

#Mask上加噪

在MASK上加噪声，然后和图像一起提特征，学习去噪，和DiffusionDet类似，需要GT训练。

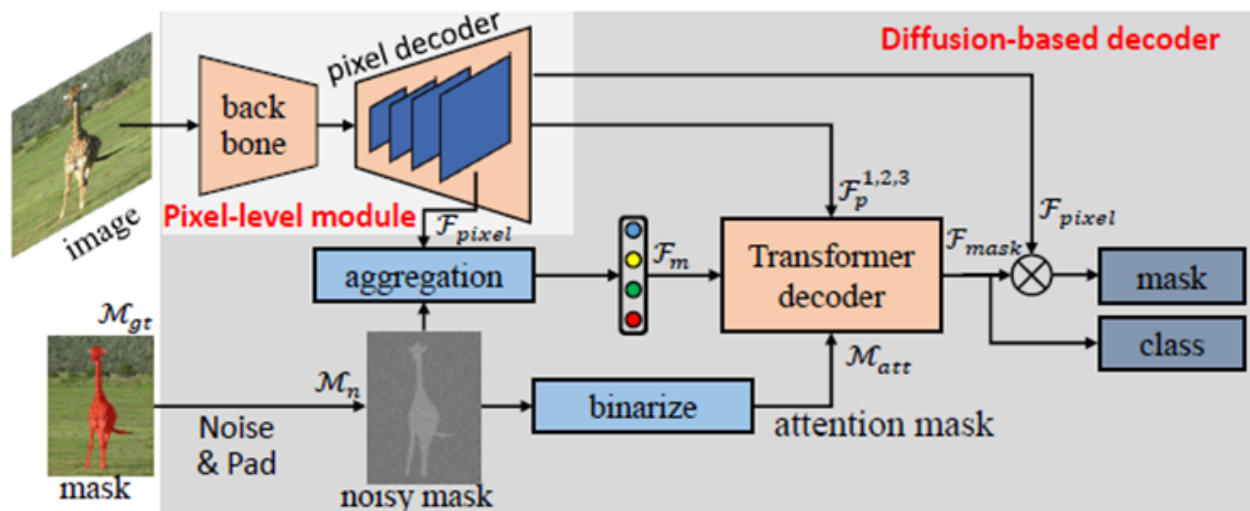


Figure 2: Overall architecture of our DFormer for universal image segmentation. We employ pixel-level module, including backbone and pixel decoder, to extract multi-scale features. During training, we perform the diffusion process by adding Gaussian noise to ground-truth mask, and employ a transformer decoder to predict mask and corresponding category from the noisy mask.

ARXIVWhat the DAAM Interpreting Stable Diffusion Using Cross Attention

#k

主要用SD交叉注意力机制

#attention得到粗略图

代码 <https://github.com/castorini/daam>



Figure 1: The original synthesized image and three DAAM maps for “monkey,” “hat,” and “walking,” from the prompt, “monkey with hat walking.”

ARXIV_Interpretable Diffusion via Information Decomposition

通过修改提示语言，互信息结果显著图更好？互信息表示比较数学化，论文很难看懂



Figure 1: We start (left) with a real image from the COCO dataset. We do a “prompt intervention” (§3.3) to generate a new image. Next we show conditional mutual information, illustrated using our pixel-wise decomposition, and attention maps for the modified word. Top row shows an image where prompt intervention has an effect, while in the bottom row it has little effect. Conditional mutual information reflects the effect of intervention while attention does not.

基于h-space的相关方法

ICLR2023_Diffusion Models Already Have A Semantic Latent Space

#h-space

提出了h-space概念承接DiffusionCLIP，但是无需微调整个扩散模型，而只需微调扩散模型里UNet架构里最中间的那一层就可以了，和Diffusion-Clip一样需要每一个性质训练一个新的模型，哪怕这个模型可迁移性较好，训练成本较低依然难以让人接受

代码 <https://kwonminki.github.io/Asyrrp/>

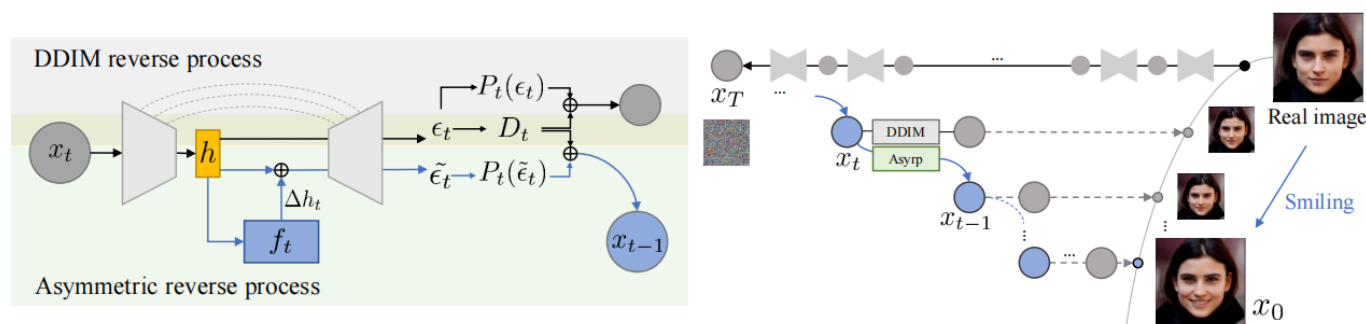


Figure 2: **Generative process of Asyrrp.** The green box on the left illustrates Asyrrp which only alters P_t while preserving D_t shared by DDIM. The right describes that Asyrrp modifies the original reverse process toward the target attribute reflecting the change in h -space.

Discovering Interpretable Directions in the Semantic Latent Space of Diffusion Models

从h-space中提取语义变换方向。大量图像上通过PCA提取全局方向，单个图像上通过特征SVD分解提取最主要语义方向，有监督图像上，特征相减得到语义方向。两个图交换h-space，背景和亮度不变，能用于目标分割？

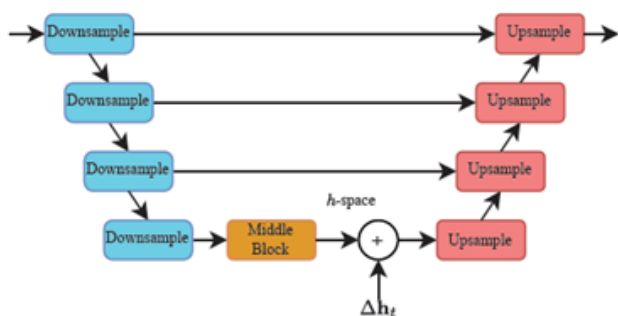


Figure 2: Illustration of h -space. In this paper we define the semantic latent space of DDMs as the activation after the deepest bottleneck layer of the U-Net.



Figure 3: Effect of swapping the bottleneck activation. Swapping h_t between two samples swaps the semantic content without affecting background and illumination.

$$\text{Image} + \gamma (\text{Image}_1 - \text{Image}_2) = \text{Image}_3$$

WACV2024_Training-free Content Injection using h-space in Diffusion Models

#风格注入

通过h-space实现内容注入，A图中注入B图的风格。

<https://curryjung.github.io/DiffStyle/>

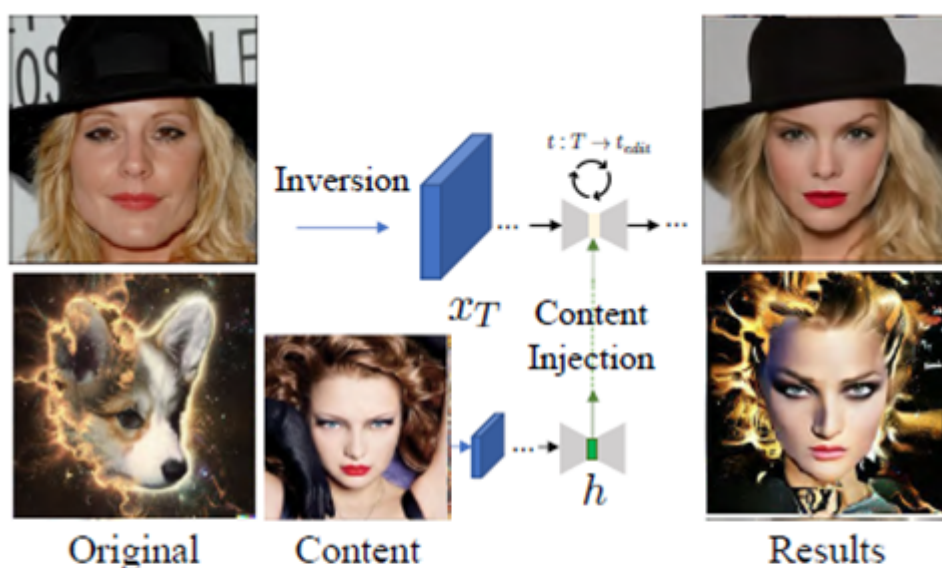


Figure 1. Overview of InjectFusion. During the content injection, the bottleneck feature map is recursively injected during the sampling process started from the inverted x_T of images. The target content is reflected in the result images while preserving the original images.

ARXIV_Unsupervised Discovery of Interpretable Directions in h-space of Pre-trained Diffusion Models

基于h-space做语义方向和幅度调整，学习一个重建器判断语义方向和幅度调整值（用GAN实现?）。

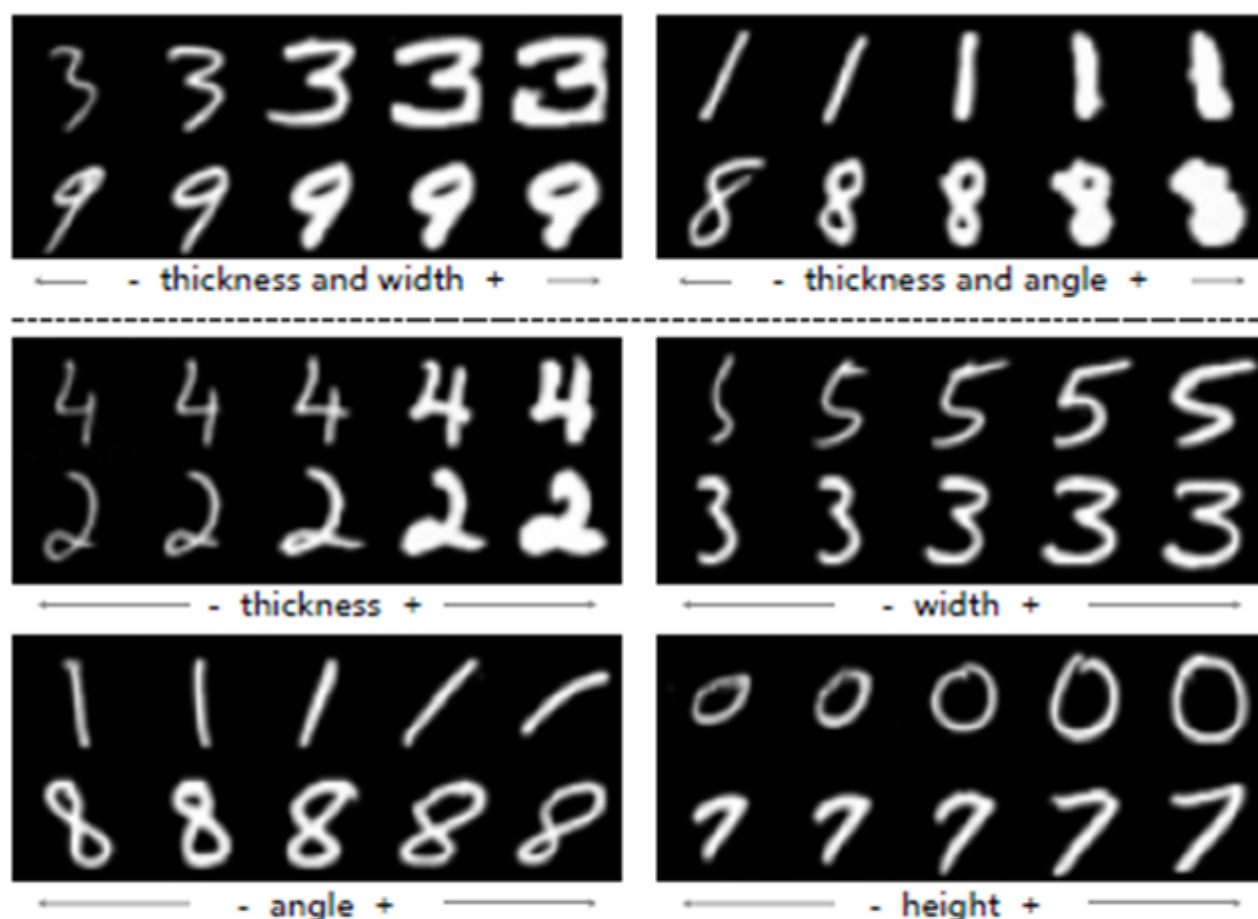


Figure 1. Interpretable directions discovered by our method in h-space of pre-trained DMs for MNIST. Each example contains five samples generated on the same discovered direction, but with different magnitudes. All attributes are manually interpreted.